

3. Variations of Brownian Motion

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All proofs from Stochastic Processes by Sheldon Ross

3.1: Absorbed Brownian Motion

An absorbed Brownian motion behaves exactly like a standard Brownian motion until it hits a specific barrier a . Once it reaches a , the process is absorbed and remains fixed at that value forever.

Definition: Let $\tau_a = \inf\{t \geq 0 : W(t) = a\}$. The absorbed process $X(t)$ is:

$$X(t) = \begin{cases} W(t) & \text{if } t < \tau_a \\ a & \text{if } t \geq \tau_a, \end{cases}$$

3.1: Absorbed Brownian Motion (Simulation)

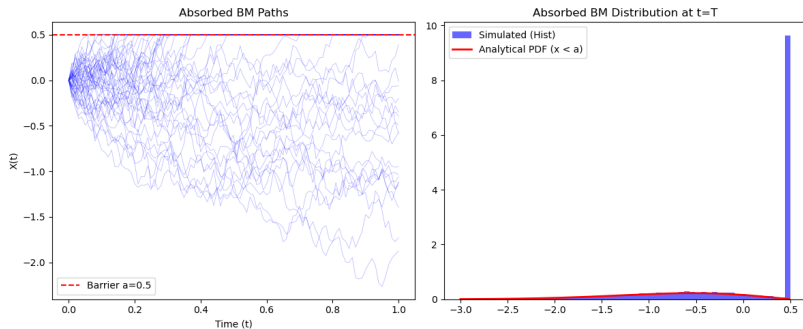


Figure: Paths and distribution of Absorbed Brownian Motion at barrier $a = 0.5$

3.1: Derivation of Absorbed BM PDF

Let $W(t)$ be a standard Brownian motion and $a > 0$ be an absorbing barrier. The stopping time is

$$\tau_a = \inf\{t > 0 : W(t) = a\}$$

For the absorbed process $X(t)$, we want to find the CDF for $x < a$:

$$P(X(t) \leq x) = P(W(t) \leq x, \tau_a > t)$$

Using the Law of Total Probability:

$$P(W(t) \leq x) = P(W(t) \leq x, \tau_a > t) + P(W(t) \leq x, \tau_a \leq t)$$

By symmetry, in standard brownian motion, paths can hit $x < a$ or $a + (a - x) = 2a - x$ at time $t \geq \tau_a$ with equal probability (going up or down the same amount has equal probability, this amount is $a - x$).

$$P(W(t) \leq x, \tau_a \leq t) = P(W(t) \geq 2a - x, \tau_a \leq t)$$

3.1: Derivation of Absorbed BM PDF

Now observe that $W(t) > 2a - x$ implies $W(t) > a$ (since $x < a$). Thus, this means that $t \geq \tau_a$ and so the probability can be written as:

$$P(W(t) \geq 2a - x, \tau_a \leq t) = P(W(t) \geq 2a - x)$$

Substituting the probability back into the CDF equation:

$$P(X(t) \leq x) = P(W(t) \leq x) - P(W(t) \geq 2a - x)$$

Using the standard normal CDF Φ :

$$F_{X(t)}(x) = \Phi\left(\frac{x}{\sqrt{t}}\right) - \left(1 - \Phi\left(\frac{2a - x}{\sqrt{t}}\right)\right)$$

Since the normal distribution is symmetric, $1 - \Phi(z) = \Phi(-z)$:

$$F_{X(t)}(x) = \Phi\left(\frac{x}{\sqrt{t}}\right) - \Phi\left(\frac{x - 2a}{\sqrt{t}}\right)$$

Differentiating with respect to x gives the PDF for $x < a$:

$$f_{X(t)}(x) = \frac{1}{\sqrt{2\pi t}} \left(e^{-\frac{x^2}{2t}} - e^{-\frac{(x-2a)^2}{2t}} \right)$$

3.1: Hitting Time Probability in Standard BM

Now consider standard Brownian motion $W(t)$. The CDF of the hitting time τ_a is $P(\tau_a \leq t)$.

$$P(\tau_a \leq t) = P(W(t) \geq a, \tau_a \leq t) + P(W(t) < a, \tau_a \leq t)$$

By symmetry,

$$P(W(t) < a, \tau_a \leq t) = P(W(t) \geq a, \tau_a \leq t) = P(W(t) \geq a).$$

Thus:

$$P(\tau_a \leq t) = 2P(W(t) \geq a) = 2 \left(1 - \Phi \left(\frac{a}{\sqrt{t}} \right) \right)$$

To find the probability of hitting a in finite time, we take the limit as $t \rightarrow \infty$:

$$\lim_{t \rightarrow \infty} P(\tau_a \leq t) = 2(1 - \Phi(0)) = 2(1 - 0.5) = 1$$

Therefore, standard Brownian motion hits any value a in finite time with probability 1.

3.1: Expected Hitting Time is Infinite

To find the expected hitting time $\mathbb{E}[\tau_a]$, we first find the PDF of τ_a by differentiating its CDF with respect to t :

$$f_{\tau_a}(t) = \frac{d}{dt} \left[2 \int_{a/\sqrt{t}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz \right] = \frac{a}{\sqrt{2\pi} t^{3/2}} e^{-\frac{a^2}{2t}}$$

As $t \rightarrow \infty$, the integrand behaves like $t^{-1/2}$, which diverges (proof in next slide). Hence:

$$\mathbb{E}[\tau_a] = \infty$$

Because $P(\tau_a < \infty) = 1$ and $\mathbb{E}[\tau_a] = \infty$ for any arbitrary state a , every state in a standard Brownian motion is **null recurrent**.

3.1: Expected Hitting Time is Infinite

The expected value is:

$$\begin{aligned}\mathbb{E}[\tau_a] &= \int_0^\infty P(\tau_a > t) dt \\&= \int_0^\infty \left(1 - \frac{2}{\sqrt{2\pi}} \int_{a/\sqrt{t}}^\infty e^{-y^2/2} dy \right) dt \\&= \frac{2}{\sqrt{2\pi}} \int_0^\infty \int_0^{a/\sqrt{t}} e^{-y^2/2} dy dt \\&= \frac{2}{\sqrt{2\pi}} \int_0^\infty \int_0^{a^2/y^2} dt e^{-y^2/2} dy \\&= \frac{2a^2}{\sqrt{2\pi}} \int_0^\infty \frac{1}{y^2} e^{-y^2/2} dy \\&\geq \frac{2a^2 e^{-1/2}}{\sqrt{2\pi}} \int_0^1 \frac{1}{y^2} dy \\&= \infty.\end{aligned}$$

3.2: Reflected Brownian Motion at Origin

Reflected Brownian motion behaves like a standard Brownian motion but is constrained to be non-negative. Whenever the process hits 0, it bounces back into the positive domain.

Definition: The standard reflected Brownian motion is the absolute value of a standard Brownian motion:

$$Z(t) = |W(t)|$$

3.2: Reflected Brownian Motion (Simulation)

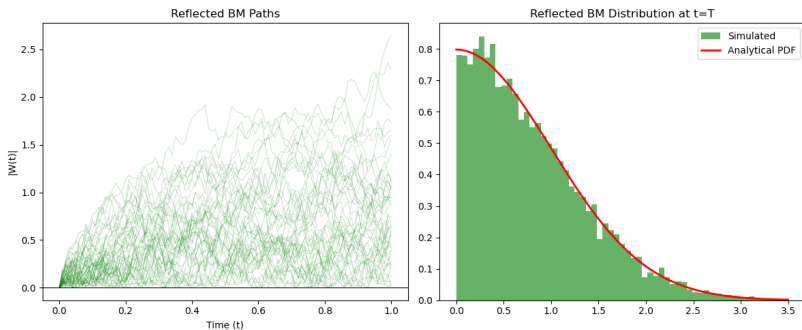


Figure: Paths and distribution of Reflected Brownian Motion

3.2: Distribution of Reflected Brownian Motion

The distribution of $Z(t)$ is obtained as:

For $y > 0$,

$$\begin{aligned}P(Z(t) \leq y) &= P(-y \leq W(t) \leq y) \\&= P(W(t) \leq y) - P(W(t) \leq -y) \\&= 2P(W(t) \leq y) - 1 \\&= \frac{2}{\sqrt{2\pi t}} \int_{-\infty}^y e^{-x^2/2t} dx - 1,\end{aligned}$$

($P(W(t) \leq -y) = 1 - P(W(t) \leq y)$, due to $W(t)$ having gaussian distribution)

- ▶ Expected value: $\mathbb{E}[Z(t)] = \sqrt{\frac{2t}{\pi}}$
- ▶ Variance: $\text{Var}(Z(t)) = \left(1 - \frac{2}{\pi}\right) t$

3.3: Geometric Brownian Motion

If $\{W(t), t \geq 0\}$ is a standard Brownian motion, then the process $\{Y(t), t \geq 0\}$, defined by

$$Y(t) = e^{W(t)}, \quad t \geq 0,$$

is called **geometric Brownian motion**.

3.3: Geometric Brownian Motion (Simulation)

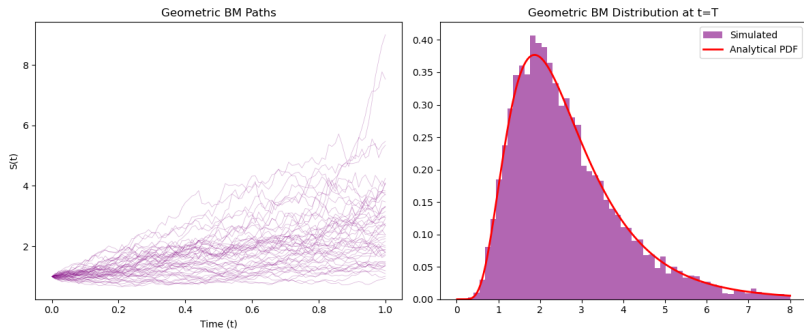


Figure: Paths and Log-Normal distribution of GBM

3.3: Geometric Brownian Motion: Definition and Moments

Since $W(t)$ is normal with mean 0 and variance t , its MGF is given by:

$$\mathbb{E}[e^{sW(t)}] = e^{ts^2/2}$$

Using this, we can compute the mean and variance of $Y(t)$:

$$\begin{aligned}\mathbb{E}[Y(t)] &= \mathbb{E}[e^{W(t)}] = e^{t/2} \\ \text{Var}(Y(t)) &= \mathbb{E}[Y^2(t)] - (\mathbb{E}[Y(t)])^2 \\ &= \mathbb{E}[e^{2W(t)}] - e^t \\ &= e^{2t} - e^t.\end{aligned}$$

3.3: GBM Example: Modelling stock prices

Using discrete real-world observations and assumptions (like stock prices):

- ▶ Let $Y(n)$ be a stock price at day n .
- ▶ Assume the daily **percentage returns**, $X_i = \frac{Y(i)}{Y(i-1)}$, are i.i.d.

The price at time n is the product of all previous returns:

$$Y(n) = Y(0)X_1X_2\cdots X_n$$

Taking the logarithm turns the product into a sum of i.i.d. variables:

$$\log Y(n) = \log Y(0) + \sum_{i=1}^n \log X_i$$

By the **Central Limit Theorem**, this sum of i.i.d. variables converges to a Brownian motion as the time steps become continuous. Thus, $Y(t)$ naturally becomes a Geometric Brownian Motion.

3.3: Discrete Prices and Multiplicative Returns

Before jumping to continuous calculus, we observe real-world markets in discrete time steps (e.g., daily closing prices).

- ▶ Let $Y(n)$ be the price of an asset at day n .
- ▶ Let $X_i = \frac{Y(i)}{Y(i-1)}$ be the gross return on day i .
- ▶ We assume these daily returns X_i are **independent and identically distributed (i.i.d.)**.

The price at day n is simply the initial price multiplied by all subsequent daily returns:

$$Y(n) = Y(0)X_1X_2 \cdots X_n$$

3.3: Discrete Prices and Multiplicative Returns

Taking the natural logarithm transforms the product into a sum:

$$\log Y(n) = \log Y(0) + \sum_{i=1}^n \log X_i$$

- ▶ Let $Z_i = \log X_i$. Since the X_i are i.i.d., the Z_i terms are also i.i.d.
- ▶ Let the mean be $\mathbb{E}[Z_i] = \mu$ and variance be $\text{Var}(Z_i) = \sigma^2$.
- ▶ We define our sum: $S_n = \sum_{i=1}^n Z_i$.

The Central Limit Theorem (CLT) applies to our sum S_n , by appropriate scaling.

Subtract its mean ($n\mu$) and divide by its standard deviation ($\sigma\sqrt{n}$):

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{as } n \rightarrow \infty$$

Thus S_n becomes a gaussian random variable with mean $n\mu$ and variance $n\sigma^2$.

$$S_n \sim \mathcal{N}(n\mu, n\sigma^2)$$

3.4: Integrated Brownian Motion

If $\{W(t), t \geq 0\}$ is a standard Brownian motion, the **integrated Brownian motion** $\{Y(t), t \geq 0\}$ is defined as the area under the Brownian path:

$$Y(t) = \int_0^t W(s) ds$$

Physical Intuition:

- ▶ If $W(t)$ represents the random *velocity* of a particle subject to continuous random shocks...
- ▶ ...then $Y(t)$ represents the *position* of that particle.

Smoothness: While $W(t)$ is jagged and nowhere differentiable, integration is a smoothing operation. $Y(t)$ is continuously differentiable, and its derivative is exactly $\frac{d}{dt} Y(t) = W(t)$. The path for IBM is smooth.

3.4: Integrated Brownian Motion (Simulation)

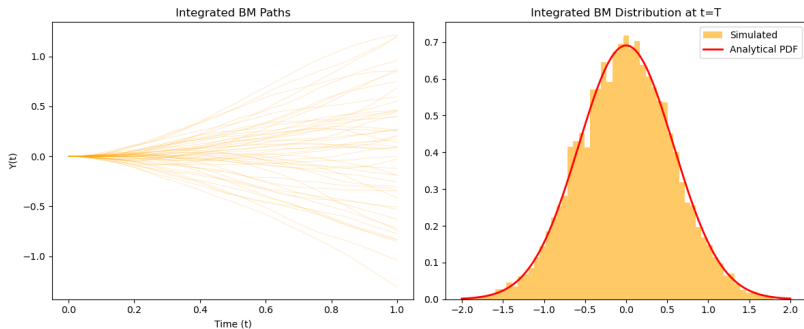


Figure: Paths and distribution of Integrated Brownian Motion

3.4: $Y(t)$ is Gaussian

The variables of a Brownian motion at any finite set of times are **jointly Gaussian**.

Consider any sequence of times $0 < t_1 < t_2 < \dots < t_n$. By the definition of Brownian motion, the *increments* are independent and normally distributed:

$$Z_1 = W(t_1) - W(0) \sim \mathcal{N}(0, t_1)$$

$$Z_2 = W(t_2) - W(t_1) \sim \mathcal{N}(0, t_2 - t_1)$$

$$\vdots$$

$$Z_n = W(t_n) - W(t_{n-1}) \sim \mathcal{N}(0, t_n - t_{n-1})$$

Because the increments Z_i are independent Gaussians, the vector $\mathbf{Z} = [Z_1, Z_2, \dots, Z_n]^T$ is trivially a **jointly Gaussian vector**.

3.4: $Y(t)$ is Gaussian

We can express the actual Brownian motion values $W(t_k)$ as a cumulative sum of these independent Gaussian increments:

$$W(t_1) = Z_1$$

$$W(t_2) = Z_1 + Z_2$$

$$\vdots$$

$$W(t_n) = Z_1 + Z_2 + \cdots + Z_n$$

Property: Any linear transformation of a jointly Gaussian vector results in another jointly Gaussian vector.

Since $\mathbf{W} = [W(t_1), \dots, W(t_n)]^T$ is just a linear combination of $\mathbf{Z}$, we conclude that $W(t)$ **at any set of time instants is jointly Gaussian**.

3.4: $Y(t)$ is Gaussian

Using the definition of Riemann integral as a limit of sums over small time steps Δs :

$$Y(t) \approx \sum_{i=1}^n W(s_i) \Delta s$$

- ▶ From our previous proof, the variables $(W(s_1), W(s_2), \dots, W(s_n))$ are jointly Gaussian.
- ▶ A finite linear combination of jointly Gaussian variables is strictly Gaussian.
- ▶ The limit of a sequence of Gaussian random variables (as $\Delta s \rightarrow 0$) is also Gaussian.

Therefore, $Y(t)$ is a normally distributed random variable. We just need its parameters.

3.4: Parameters of IBM: Expected Value

Since integration is a linear operator, we can pass the expectation directly inside the integral, indirectly applying linearity of expectation to variables in the Riemann integral
The mean of the integrated Brownian motion is:

$$\begin{aligned}\mathbb{E}[Y(t)] &= \mathbb{E}\left[\int_0^t W(s) ds\right] \\ &= \int_0^t \mathbb{E}[W(s)] ds\end{aligned}$$

Because standard Brownian motion has a mean of zero at all times ($\mathbb{E}[W(s)] = 0$):

$$\mathbb{E}[Y(t)] = \int_0^t 0 ds = 0$$

3.4: Parameters of IBM: Variance

To find the variance, we calculate $\mathbb{E}[Y(t)^2]$ since the mean is zero. We square the integral by writing it as a double integral with dummy variables u and v :

$$\begin{aligned}\text{Var}(Y(t)) &= \mathbb{E} \left[\left(\int_0^t W(s) ds \right)^2 \right] \\ &= \mathbb{E} \left[\int_0^t W(u) du \int_0^t W(v) dv \right] \\ &= \mathbb{E} \left[\int_0^t \int_0^t W(u) W(v) du dv \right]\end{aligned}$$

Passing the expectation inside the double integral:

$$\text{Var}(Y(t)) = \int_0^t \int_0^t \mathbb{E}[W(u)W(v)] du dv$$

3.4: Covariance of Standard Brownian Motion

To evaluate the double integral for the variance, we need the covariance of standard Brownian motion: $\mathbb{E}[W(u)W(v)]$. Let's derive it.

Assume without loss of generality that $u \leq v$. We can rewrite the value of the Brownian motion at the later time v as its value at the earlier time u plus the increment that occurred between u and v :

$$W(v) = W(u) + (W(v) - W(u))$$

Substitute this expanded form of $W(v)$ into the expectation:

$$\begin{aligned}\mathbb{E}[W(u)W(v)] &= \mathbb{E}[W(u) \cdot (W(u) + W(v) - W(u))]\cr &= \mathbb{E}[W(u)^2 + W(u)(W(v) - W(u))]\end{aligned}$$

3.4: Covariance of Standard Brownian Motion

Using the linearity of expectation, we split the expression into two terms, and using $W(0) = 0$:

$$\mathbb{E}[W(u)W(v)] = \mathbb{E}[W(u)^2] + \mathbb{E}[(W(u) - W(0))(W(v) - W(u))]$$

Now, we analyze the two terms:

1. **First Term:** Since $W(u) \sim \mathcal{N}(0, u)$, its second moment (which is its variance) is exactly $\mathbb{E}[W(u)^2] = u$.
2. **Second Term:** Because Brownian motion has *independent increments*, the increment $W(u) - W(0)$ is independent of the non overlapping future increment $W(v) - W(u)$. Thus:

$$\mathbb{E}[W(u) - W(0)] \cdot \mathbb{E}[W(v) - W(u)] = 0 \cdot 0 = 0$$

Adding them yields $\mathbb{E}[W(u)W(v)] = u + 0 = u$. Similarly, if $v \leq u$ instead, the result would be v . Therefore, the covariance is always $\min(u, v)$.

3.4: Parameters of IBM: Covariance

Since $\{Y(t), t \geq 0\}$ is a Gaussian process, its distribution is fully characterized by its mean value and covariance function.

For $s \leq t$, we compute the covariance:

$$\begin{aligned}\text{Cov}[Y(s), Y(t)] &= \mathbb{E}[Y(s)Y(t)] \\ &= \mathbb{E}\left[\int_0^s W(y) dy \int_0^t W(u) du\right] \\ &= \int_0^s \int_0^t \mathbb{E}[W(y)W(u)] du dy \\ &= \int_0^s \int_0^t \min(y, u) du dy\end{aligned}$$

3.4: Parameters of IBM: Covariance

To evaluate $\int_0^s \int_0^t \min(y, u) du dy$ for $s \leq t$, we split the inner integral at $u = y$:

$$\begin{aligned}\text{Cov}[Y(s), Y(t)] &= \int_0^s \left(\int_0^y u du + \int_y^t y du \right) dy \\ &= \int_0^s \left(\frac{y^2}{2} + y(t - y) \right) dy \\ &= \int_0^s \left(yt - \frac{y^2}{2} \right) dy \\ &= t \frac{s^2}{2} - \frac{s^3}{6} \\ &= s^2 \left(\frac{t}{2} - \frac{s}{6} \right)\end{aligned}$$

Note: Setting $s = t$ immediately recovers the variance $t^3/3$.

Conclusion: For any time t , $Y(t) \sim \mathcal{N}\left(0, \frac{t^3}{3}\right)$.

3.4: Markov Process ($Y(t)$, $W(t)$)

While $Y(t)$ alone is not a Markov process, the vector process $\{(Y(t), W(t)), t \geq 0\}$ is Markov. Because they are jointly normal, their joint distribution is defined by their cross-covariance.

We analyze the difference over a small step h :

$$\begin{aligned}\text{Cov}(Y(t), Y(t) - Y(t - h)) &= \text{Cov}(Y(t), Y(t)) - \text{Cov}(Y(t), Y(t - h)) \\ &= \frac{t^3}{3} - (t - h)^2 \left[\frac{t}{2} - \frac{t - h}{6} \right] \\ &= \frac{t^2 h}{2} + o(h)\end{aligned}$$

Alternatively, expressing the difference as an integral:

$$\begin{aligned}\text{Cov}(Y(t), Y(t) - Y(t - h)) &= \text{Cov} \left(Y(t), \int_{t-h}^t W(s) ds \right) \\ &= \text{Cov}(Y(t), hW(t) + o(h)) \\ &= h\text{Cov}(Y(t), W(t)) + o(h)\end{aligned}$$

3.4: The Bivariate Normal Distribution

Equating the two expressions for the covariance over step h :

$$h\text{Cov}(Y(t), W(t)) \approx \frac{t^2 h}{2}$$

Dividing by h and taking the limit as $h \rightarrow 0$ gives the cross-covariance:

$$\text{Cov}(Y(t), W(t)) = \frac{t^2}{2}$$

Hence, the joint state of the integrated process and the Brownian motion, $(Y(t), W(t))$, has a bivariate normal distribution with:

$$\mathbb{E}[Y(t)] = \mathbb{E}[W(t)] = 0$$

$$\text{Var}(Y(t)) = \frac{t^3}{3}, \quad \text{Var}(W(t)) = t$$

$$\text{Cov}(Y(t), W(t)) = \frac{t^2}{2}$$

3.4: Exponential Integrated Brownian Motion

Another form of integrated Brownian motion arises if we assume the *percent rate of change* of a price follows a Brownian motion process (similar to what we did in GBM).

If $P(t)$ denotes the price at time t , then:

$$\frac{d}{dt}P(t) = W(t)P(t)$$

Solving this differential equation yields:

$$P(t) = P(0) \exp \left\{ \int_0^t W(s) ds \right\} = P(0)e^{Y(t)}$$

Taking $P(0) = 1$, we have $P(t) = e^{Y(t)}$. Since $Y(t)$ is normal with mean 0 and variance $t^3/3$, $P(t)$ is log-normally distributed.

$$\mathbb{E}[P(t)] = \mathbb{E} \left[e^{Y(t)} \right] = \exp \left\{ \frac{t^3}{6} \right\}$$

The above comes from the $M_Y(1)$ where M_Y is the MGF of $Y(t)$.

3.5: Brownian Motion with Drift

A Brownian motion with drift represents a standard Brownian motion that is shifted by a constant trend over time.

Definition: The process $X(t)$ is defined as:

$$X(t) = \mu t + \sigma W(t)$$

Where μ is the constant drift rate and σ is the volatility. At time t , $X(t)$ is normally distributed with mean μt and variance $\sigma^2 t$.

3.5: Brownian Motion with Drift (Simulation)

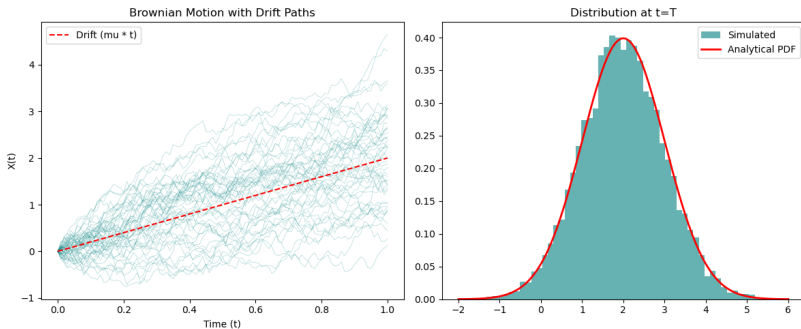


Figure: Paths and distribution of Brownian Motion with Drift

3.5: Optimal Stopping Time

Let $X(t)$ be a Brownian motion with drift μ and variance $\sigma^2 t$.

We want to compute the probability that the process hits an upper barrier $A > 0$ before a lower barrier $-B < 0$.

Let $P(x)$ denote this probability, conditioned on starting at a position $x \in (-B, A)$. So $X(0)$ need not be 0:

$$P(x) = P(X(t) \text{ hits } A \text{ before } -B \mid X(0) = x)$$

To find $P(x)$, we will observe the process over a microscopic time interval $[0, h]$. Let $Y = X(h) - X(0)$ be the random step taken during this time.

3.5: Optimal Stopping Time

Let H be the event that the process hits either barrier (A or $-B$) during the microscopic time interval $[0, h]$. Let Win be the event that the process hits A before it hits $-B$.

By the Law of Total Probability, we can split $P(x)$ into two scenarios:

$$\begin{aligned} P(x) &= P(\text{Win} \mid H)P(H) \\ &\quad + P(\text{Win} \mid H^c)P(H^c) \end{aligned}$$

Because Brownian paths are continuous and x is strictly inside $(-B, A)$, moving a macroscopic distance to a boundary in a infinitesimal time h is small.

Mathematically, $P(H)$ shrinks to zero faster than h , so $P(H) = o(h)$.

3.5: Optimal Stopping Time

Using $P(H) = o(h)$,

$$P(x) = P(\text{Win} \mid H) \cdot o(h) + P(\text{Win} \mid H^c)(1 - o(h))$$

$$P(x) = P(\text{Win} \mid H^c) + o(h)$$

(Note that $o(h) \times \text{bounded} = o(h)$, here the difference of probabilities is bounded between -1 and 1)

This tells us that for an infinitesimally small step h , we can safely assume the process has not yet hit a boundary.

3.5: Optimal Stopping Time

Markov property dictates that the future evolution of the process is dependent only on the present:

$$\mathbb{P}(\text{Future} \mid \text{Present, Past}) = \mathbb{P}(\text{Future} \mid \text{Present})$$

$$\mathbb{P}(W \mid X(h) = x + Y, H^C, X(0) = x) = \mathbb{P}(W \mid X(h) = x + Y)$$

Because Brownian motion is **time-homogeneous**, the probability of winning starting from position $x + Y$ at time h is identical to starting there at time 0:

$$\mathbb{P}(W \mid X(h) = x + Y) = P(x + Y)$$

Integrating over all possible values $Y = y$ of the random step $Y = X(h) - X(0)$ yields the total expectation: $\mathbb{E}[P(x + Y)]$.

3.5: Optimal Stopping Time

We now proceed by taking a Taylor series expansion of $P(x + Y)$ about x :

$$P(x + Y) = P(x) + P'(x)Y + \frac{P''(x)}{2}Y^2 + \dots$$

Substitute this expansion into our expectation equation:

$$P(x) = \mathbb{E} \left[P(x) + P'(x)Y + \frac{P''(x)}{2}Y^2 \right] + o(h)$$

Since $Y \sim \mathcal{N}(\mu h, h)$, we know its moments:

- ▶ $\mathbb{E}[Y] = \mu h$
- ▶ $\mathbb{E}[Y^2] = \text{Var}(Y) + (\mathbb{E}[Y])^2 = \sigma^2 h + \mu^2 h^2 = h + o(h)$

(Higher-order terms like Y^3 yield expectations that are $o(h)$).

3.5: Optimal Stopping Time

Applying the expectations to our expansion:

$$P(x) = P(x) + P'(x)\mu h + \frac{P''(x)}{2}\sigma^2 h + o(h)$$

Subtract $P(x)$ from both sides, divide the entire equation by h , and isolate the terms:

$$P'(x)\mu + \frac{P''(x)}{2}\sigma^2 = \frac{o(h)}{h}$$

Letting the time step $h \rightarrow 0$, the $\frac{o(h)}{h}$ term vanishes by definition. This leaves us with a second-order linear differential equation:

$$P'(x)\mu + \frac{P''(x)}{2}\sigma^2 = 0$$

3.5: Optimal Stopping Time

Integrating once yields $e^{2(\mu/\sigma^2)x}P'(x) = c_1$. Integrating a second time gives the general solution:

$$P(x) = C_1 + C_2 e^{-2(\mu/\sigma^2)x}$$

We use the boundary conditions $P(A) = 1$ and $P(-B) = 0$ to solve for C_1 and C_2 :

$$C_1 + C_2 e^{-2(\mu/\sigma^2)A} = 1$$

$$C_1 + C_2 e^{2(\mu/\sigma^2)B} = 0$$

Solving this linear system yields:

$$C_1 = \frac{e^{2(\mu/\sigma^2)B}}{e^{2(\mu/\sigma^2)B} - e^{-2(\mu/\sigma^2)A}}, \quad C_2 = \frac{-1}{e^{2(\mu/\sigma^2)B} - e^{-2(\mu/\sigma^2)A}}$$

Substituting C_1 and C_2 back gives the probability from any starting point x :

$$P(x) = \frac{e^{2(\mu/\sigma^2)B} - e^{-2(\mu/\sigma^2)x}}{e^{2(\mu/\sigma^2)B} - e^{-2(\mu/\sigma^2)A}}$$